

System limits

The sizes, counts and depths compiled into SD Core for Windows, and which of them a site can change.

Two kinds of number appear on this page and they behave differently.

A **compiled limit** is fixed in the build. Changing it means rebuilding SD, and several would change on-disk file formats, so in practice they are ceilings.

A **configured limit** is a value in `sd.conf` with a default. A site can raise or lower it, within a range the compiled limits set. Those are covered in full under *Configuration* in this set; this page gives the number in force and says which parameter controls it.

This document is separate so that it can be withheld. It links to nothing outside the administrator set. Where a page in another set is worth naming, it is named in words.

Every figure here was read from the sources that define it, or from a running W1.0-0 system, and not from documentation of another product.

Data

Maximum string or record size	1,073,741,822 bytes	compiled
Maximum record id length	63 by default, up to 255	MAXIDLEN
Fields per record	limited only by record size	—
Values per field, subvalues per value	limited only by record size	—

The record size limit is just under 1 GB rather than 2 GB. It is the maximum size of any single string SD will build, so it applies to a record, to a concatenation, and to anything a program assembles in memory.

A record id may be longer than 63 characters only if `MAXIDLEN` is raised, and the range starts at 63 rather than 0 — a lower value is refused at start-up rather than accepted. The ceiling of 255 is compiled, and the comment in the source notes that raising it would require major file-format changes.

Files

File types	two — directory and dynamic	compiled
Group size for a dynamic file	2 KB by default, maximum 8	<code>GRPSIZE</code>
Open files across all sessions	80	<code>NUMFILES</code>
Maximum pathname length	255	compiled
Sort keys in one query	32	compiled

Group size is expressed in 1 KB units, so the compiled maximum of 8 is 8 KB.

Sessions and locks

Concurrent sessions	20	<code>NUMUSERS</code>
Record locks across all sessions	100	<code>NUMLOCKS</code>
File locks	one per file	compiled
Windows account name	32 characters	compiled
SD account name	32 characters	compiled

`NUMUSERS` sizes the user table in shared memory, so it is read once when SD starts and cannot be changed in a running system.

Programs

Subroutine call depth	10,000	<code>MAXCALL</code> , range 10 to 1,000,000
Catalogued name length	63 characters	compiled
Matrix dimensions	two	compiled

Compiled programs held in memory	no limit by default	<code>OBJECTS</code>
Memory those programs may occupy	no limit by default	<code>OBJMEM</code> , in KB

Select lists

Numbered select lists	13 , numbered 0 to 12	compiled
Items in a select list	limited by disk	—

Thirteen is the number a session has available at once, not a limit on how many lists can be saved. `save.list` writes to a file and is limited only by space.

Terminals

Terminal definitions shipped	63 source definitions	—
Terminal names they compile to	100	—
Default page size	120 × 36	—
Minimum page size	10 lines × 20 columns	compiled
Default printer page	80 × 66	<code>LPTRWIDE</code> , <code>LPTRHIGH</code>

The extra terminal names are aliases and variants that share a definition with their base name.

What each account tier gets

These are inventories rather than limits, and they are the numbers most often misquoted, so they are given here with what they count.

	VERBS	VOC RECORDS
Standard	82	355

	VERBS	VOC RECORDS
Programmer	124	397
Administrator	147	420

The two columns count different things. A VOC record may be a verb, a keyword, a file pointer, a sentence or a paragraph; only some are verbs. A reader asking "how many commands can this account type?" wants the first column. A tool comparing two accounts with `count voc` sees the second.

Both are computed from the shipped VOC rather than maintained by hand: the verb figures come from the same two tier lists the account-creation code reads, and the record figures are what `count voc` reports in a freshly created account.

Configuration

Parameters accepted in <code>sd.conf</code>	52
Parameters the <code>config</code> verb displays	43
Parameters readable with <code>config()</code>	49
Parameters settable for the current session	28
Parameters accepted that do nothing	6

The six inert parameters, and why each is inert, are listed under *Configuration* in this set. `NETFILES` and `CREATUSR` are the two a site is most likely to have in an existing `sd.conf`.